Calculus & Algebra

angles in radians · log means the natural log · **priority**: must be automatic · recurring · in scope · context

1 · Algebra

1.1 Quadratics

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \qquad \Delta = b^2 - 4ac$$

Pf. Complete the square: $a\left(x + \frac{b}{2a}\right)^2 = \frac{\Delta}{4a}$; take roots.

Meaning: the discriminant decides the number of real roots before you solve anything.

$$\Delta > 0 \Rightarrow 2 \text{ real}, \qquad \Delta = 0 \Rightarrow 1, \qquad \Delta < 0 \Rightarrow 0$$

1.2 Factorisation and logs

$$a^2 - b^2 = (a - b)(a + b), \qquad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$\log(xy) = \log x + \log y, \qquad \log \tfrac{x}{y} = \log x - \log y, \qquad \log x^n = n \log x$$

Trap. $\log(x + y) \neq \log x + \log y$ — the most common algebra slip in an exam.

1.3 Series

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} \ (|r| < 1), \qquad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

2 · Calculus

2.1 Derivatives

$$(fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$\big(f(g(x))\big)' = f'(g(x))\, g'(x)$$

Meaning: every differentiation rule is the chain rule plus one algebraic identity.

2.2 Integrals

$$\int u\, dv = uv - \int v\, du, \qquad \int_a^b f = \big[F\big]_a^b$$

$$\int \frac{dx}{x} = \ln|x| + C, \qquad \int e^{ax} dx = \frac{e^{ax}}{a} + C$$

2.3 Taylor and Taylor error

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k + R_n(x)$$

$$|R_n(x)| \leq \frac{M\,|x-a|^{n+1}}{(n+1)!}, \qquad M = \max_{\xi} |f^{(n+1)}(\xi)|$$

Pf. Repeated integration by parts leaves the remainder in Lagrange form.

2.4 Common limits

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x \to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$

3 · Linear algebra

3.1 Determinants and inverses

$$\det(A) = \prod_i \lambda_i, \qquad A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad \det(AB) = \det A \det B$$

3.2 Eigenvalues

$$Av = \lambda v, \qquad \det(A - \lambda I) = 0$$

Meaning: the characteristic polynomial turns an eigenproblem into root finding.

3.3 Useful identities

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2$$

$A^{\mathsf{T}}A$ is symmetric and positive semi-definite

Trap. $\langle x, y\rangle = 0$ means orthogonal, **not** independent — check the dimension first.

3.4 Boundary cases

Quantity	Zero iff	Maximum iff
$\det A$	A singular	A orthogonal
$\|Ax\|$	$x \in \ker A$	x along the top singular vector
$\operatorname{rank} A$	$A = 0$	A invertible

4 · Trigonometry

4.1 Identities

$$\sin^2\theta + \cos^2\theta = 1, \qquad \sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$$

Meaning: every double-angle identity follows from the addition formula with $\beta = \alpha$.

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

4.2 Triangle rules

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Trap. The law of cosines reduces to Pythagoras only when $C = 90°$; do not drop the cosine term because the triangle "looks right-angled".

4.3 Inverse and hyperbolic

$$\arctan x + \arctan\tfrac{1}{x} = \tfrac{\pi}{2} \ (x > 0)$$

$$\cosh^2 x - \sinh^2 x = 1, \qquad \frac{d}{dx}\tanh x = 1 - \tanh^2 x$$

5 · Probability and combinatorics

5.1 Bayes and total probability

$$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}, \qquad P(B) = \sum_i P(B \mid A_i)P(A_i)$$

Pf. Write $P(A \cap B)$ both ways and divide by $P(B)$.

Meaning: the denominator is the normaliser; everything else is the definition of conditional probability applied twice.

5.2 Expectation and variance

$$\mathbb{E}[X] = \sum_x x\, P(x), \qquad \operatorname{Var} X = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b, \qquad \operatorname{Var}(aX + b) = a^2 \operatorname{Var} X$$

5.3 Counting

$$P(n, k) = \frac{n!}{(n-k)!}, \qquad C(n, k) = \frac{n!}{k!\,(n-k)!}$$

$$(x + y)^n = \sum_{k=0}^{n} C(n, k) x^k y^{n-k}$$

Trap. $C(n, k) = C(n, n-k)$ but $P(n, k) \neq P(n, n-k)$ — order matters in one and not the other.

5.4 Useful bounds

$$P(A \cup B) \leq P(A) + P(B), \qquad P(A \cap B) \geq P(A) + P(B) - 1$$

$$\big|\mathbb{E}[XY]\big| \leq \sqrt{\mathbb{E}[X^2]\,\mathbb{E}[Y^2]}$$